

Rice Grain Classification for Food Fraud Detection Using Linear Models and Neural Networks

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Dataset:

<https://www.kaggle.com/datasets/muratkokludataset/rice-image-dataset>

Of the many varieties of rice produced in West Africa, none has become as popular or preferred as jasmine rice. With such a high demand for this variety, countries like Ghana have experienced an increasing rate of food fraud through substitution of jasmine with another rice variety called agra. This adulteration negatively affects consumers who are willing to spend extra money for jasmine rice, and also harms businesses that don't take part in food fraud, who get out-competed. While there are some methods of determining the exact strain of a grain of rice, they involve very expensive and time-consuming techniques in a laboratory that aren't easily accessible to developing countries. In the Science Direct article, researchers investigated the application of pocket-sized NIR spectrometers at on-site agricultural environments to help buyers and sellers verify the quality and strain of their rice. These tools were non-invasive and were able to extract the chemical composition of rice grains for evaluation, making this a very promising solution. The goal for this project is to find another method of identifying rice grains using their images, by utilizing concepts explored and learned in class. We hope to produce similarly effective results in order to demonstrate possible real-life applications of convolutional neural networks, or CNNs.

The project idea is to create an image recognition model to detect different types of rice grains utilizing Convolutional Neural Networks and simple linear models to identify and classify rice varieties. This idea of evaluating models for rice grain classification came from the Springer article that talks about evaluating different neural network performance on a dataset aimed to classify different grains of rice. The project aims to distinguish between visually similar rice grains and analyze how accurately different machine learning models can perform this classification task. Through image recognition, it can be utilized to support food quality control systems and help detect mislabeled or adulterated rice products, contributing to the prevention of food fraud. One idea is to potentially convert images to grayscale, vectorize them by flattening pixel values into a one-dimensional feature vector for use in a linear classification model, and compare its performance to a CNN that leverages spatial relationships to learn features such as grain boundaries, texture, and shape.

Software

Language:

Python

Dataset:

Kaggle Rice Image Dataset

Libraries:

- **TensorFlow / Keras / PyTorch / Scikit-learn** – Used to build and train machine learning models, including Linear models and CNNs.
- **NumPy** – Used for numerical operations and array manipulation.
- **Pillow** – jpg manipulation
- **Pandas** – Used for dataset handling and preprocessing.

References

Teye, Ernest, and Charles L.Y. Amuah. “Rice Varietal Integrity and Adulteration Fraud Detection by Chemometrical Analysis of Pocket-Sized NIR Spectra Data.” *Applied Food Research*, vol. 2, no. 2, Dec. 2022, article 100218, <https://doi.org/10.1016/j.afres.2022.100218>

Shuvo, Syme Al-Azad, et al. “Image-Based Classification of Bangladeshi Rice Varieties Using Deep Convolutional Neural Networks.” *Agricultural Products Processing and Storage*, vol. 2, article 26, 19 Jan. 2026, <https://doi.org/10.1007/s44462-026-00061-9>

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